LAB Assignment - 6

Name: Ankur Vats

Reg. No: 23BCE7555

Slot: L37 + 38

1. Use MongoDB to implement the following DB operations

1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".

```
mongosh mongodb+srv://<cl
PS C:\Users\rocki> mongosh "mongodb+srv://labcluster.5gvihuq.mongodb.net/" --apiVersion 1 --username ankur23bce7555_db_user --pass
Current Mongosh Log ID: 69b13109ca81b07ee5718dc3
Connecting to:          mongodb+srv://<credentials>@labcluster.5gvihuq.mongodb.net/?appName=mongosh+2.5.5
Using MongoDB:          8.0.19 (API Version 1)
Using Mongosh:          2.5.5
mongosh 2.7.0 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

Atlas atlas-8o45zw-shard-0 [primary] test> use vehicles
switched to db vehicles
Atlas atlas-8o45zw-shard-0 [primary] vehicles> db
vehicles
```

2. Write a MongoDB query to display all the databases.

```
Atlas atlas-8o45zw-shard-0 [primary] vehicles> show dbs
sample_mflix   71.04 MiB
admin         360.00 KiB
local          92.40 GiB
```

3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.

```
Atlas atlas-8o45zw-shard-0 [primary] vehicles> db.createCollection("two_wheelers", {
...   capped: true,
...   size: 1048576,
...   max: 1000
... })
...
{ ok: 1 }
Atlas atlas-8o45zw-shard-0 [primary] vehicles> db.createCollection("four_wheelers")
{ ok: 1 }
```

4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
Atlas atlas-8o45zw-shard-0 [primary] vehicles> db.two_wheelers.insertMany([
...   {
...     bike_name: "Hero Splendor Plus",
...     model: "gear",
...     category: "100cc",
...     colors_available: ["black", "red", "blue"],
...     manufacturer: "Hero MotoCorp",
...     performance: 7,
...     timestamp: ISODate("2020-01-15T00:00:00Z"),
...     price: 65000
...   },
...   {
...     bike_name: "Honda Activa 6G",
...     model: "gearless",
...     category: "125cc",
...     colors_available: ["white", "grey", "blue"],
...     manufacturer: "Honda",
...     performance: 8,
...     timestamp: ISODate("2021-03-10T00:00:00Z"),
...     price: 75000
...   },
...   {
...     bike_name: "Bajaj Pulsar 150",
...     model: "gear",
...     category: "150cc",
...     colors_available: ["black", "red", "sport red"],
...     manufacturer: "Bajaj",
...     performance: 8,
...     timestamp: ISODate("2019-08-20T00:00:00Z"),
...     price: 95000
...   },
...   {
...     bike_name: "TVS Apache RTR 200",
...     model: "gear",
...     category: "200cc",
...     colors_available: ["white", "black", "grey"],
...     manufacturer: "TVS",
...     performance: 9,
...     timestamp: ISODate("2022-06-05T00:00:00Z"),
...     price: 135000
...   },
```

```
...   {
...     bike_name: "Suzuki Access 125",
...     model: "gearless",
...     category: "125cc",
...     colors_available: ["blue", "silver", "black"],
...     manufacturer: "Suzuki",
...     performance: 7,
...     timestamp: ISODate("2020-11-01T00:00:00Z"),
...     price: 78000
...   }
... ])
...
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b13169ca81b07ee5718dc4'),
    '1': ObjectId('69b13169ca81b07ee5718dc5'),
    '2': ObjectId('69b13169ca81b07ee5718dc6'),
    '3': ObjectId('69b13169ca81b07ee5718dc7'),
    '4': ObjectId('69b13169ca81b07ee5718dc8')
  }
}
```

5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
Atlas atlas-8o45zw-shard-0 [primary] vehicles> db.four_wheelers.insertMany([
...   {
...     vehicle_name: "Maruti Swift",
...     model: "own",
...     category: "car",
...     variants: ["VXI petrol", "ZXI petrol", "VXI diesel"],
...     manufacturer: "Maruti Suzuki",
...     performance: 8,
...     timestamp: ISODate("2018-05-10T00:00:00Z"),
...     price: 650000
...   },
...   {
...     vehicle_name: "Tata Ace",
...     model: "commercial",
...     category: "mini truck",
...     variants: ["diesel", "CNG"],
...     manufacturer: "Tata Motors",
...     performance: 7,
```

```
...     {
...       vehicle_name: "Ashok Leyland Boss",
...       model: "commercial",
...       category: "lorry",
...       variants: ["diesel"],
...       manufacturer: "Ashok Leyland",
...       performance: 9,
...       timestamp: ISODate("2019-02-20T00:00:00Z"),
...       price: 2200000
...     },
...     {
...       vehicle_name: "Volvo 9400",
...       model: "commercial",
...       category: "bus",
...       variants: ["diesel"],
...       manufacturer: "Volvo",
...       performance: 9,
...       timestamp: ISODate("2020-12-15T00:00:00Z"),
...       price: 8500000
...     },
...     {
...       vehicle_name: "Mahindra Blazo",
...       model: "commercial",
...       category: "heavy truck",
...       variants: ["diesel"],
...       manufacturer: "Mahindra",
...       performance: 8,
...       timestamp: ISODate("2021-04-25T00:00:00Z"),
...       price: 3000000
...     }
... ])
...
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b13170ca81b07ee5718dc9'),
    '1': ObjectId('69b13170ca81b07ee5718dca'),
    '2': ObjectId('69b13170ca81b07ee5718dcb'),
    '3': ObjectId('69b13170ca81b07ee5718dcc'),
    '4': ObjectId('69b13170ca81b07ee5718dcd')
  }
```

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

```
Atlas atlas-8o45zw-shard-0 [primary] vehicles> db.two_wheelers.find().pretty()
[
  {
    _id: ObjectId('69b13169ca81b07ee5718dc4'),
    bike_name: 'Hero Splendor Plus',
    model: 'gear',
    category: '100cc',
    colors_available: [ 'black', 'red', 'blue' ],
    manufacturer: 'Hero MotoCorp',
    performance: 7,
    timestamp: ISODate('2020-01-15T00:00:00.000Z'),
    price: 65000
  },
  {
    _id: ObjectId('69b13169ca81b07ee5718dc5'),
    bike_name: 'Honda Activa 6G',
    model: 'gearless',
    category: '125cc',
    colors_available: [ 'white', 'grey', 'blue' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: ISODate('2021-03-10T00:00:00.000Z'),
    price: 75000
  },
  {
    _id: ObjectId('69b13169ca81b07ee5718dc6'),
    bike_name: 'Bajaj Pulsar 150',
    model: 'gear',
    category: '150cc',
    colors_available: [ 'black', 'red', 'sport red' ],
    manufacturer: 'Bajaj',
    performance: 8,
    timestamp: ISODate('2019-08-20T00:00:00.000Z'),
    price: 95000
  },
  {
    _id: ObjectId('69b13169ca81b07ee5718dc7'),
    bike_name: 'TVS Apache RTR 200',
    model: 'gear',
    category: '200cc',
    colors_available: [ 'white', 'black', 'grey' ],
```

```
    colors_available: [ 'white', 'black', 'grey' ],
    manufacturer: 'TVS',
    performance: 9,
    timestamp: ISODate('2022-06-05T00:00:00.000Z'),
    price: 135000
  },
  {
    _id: ObjectId('69b13169ca81b07ee5718dc8'),
    bike_name: 'Suzuki Access 125',
    model: 'gearless',
    category: '125cc',
    colors_available: [ 'blue', 'silver', 'black' ],
    manufacturer: 'Suzuki',
    performance: 7,
    timestamp: ISODate('2020-11-01T00:00:00.000Z'),
    price: 78000
  }
]
Atlas atlas-8o45zw-shard-0 [primary] vehicles> db.four_wheelers.find().pretty()
[
  {
    _id: ObjectId('69b13170ca81b07ee5718dc9'),
    vehicle_name: 'Maruti Swift',
    model: 'own',
    category: 'car',
    variants: [ 'VXI petrol', 'ZXI petrol', 'VXI diesel' ],
    manufacturer: 'Maruti Suzuki',
    performance: 8,
    timestamp: ISODate('2018-05-10T00:00:00.000Z'),
    price: 650000
  },
  {
    _id: ObjectId('69b13170ca81b07ee5718dca'),
    vehicle_name: 'Tata Ace',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'diesel', 'CNG' ],
    manufacturer: 'Tata Motors',
    performance: 7,
    timestamp: ISODate('2017-09-01T00:00:00.000Z'),
    price: 550000
```

```
    price: 550000
  },
  {
    _id: ObjectId('69b13170ca81b07ee5718dcb'),
    vehicle_name: 'Ashok Leyland Boss',
    model: 'commercial',
    category: 'lorry',
    variants: [ 'diesel' ],
    manufacturer: 'Ashok Leyland',
    performance: 9,
    timestamp: ISODate('2019-02-20T00:00:00.000Z'),
    price: 2200000
  },
  {
    _id: ObjectId('69b13170ca81b07ee5718dcc'),
    vehicle_name: 'Volvo 9400',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Volvo',
    performance: 9,
    timestamp: ISODate('2020-12-15T00:00:00.000Z'),
    price: 8500000
  },
  {
    _id: ObjectId('69b13170ca81b07ee5718dcd'),
    vehicle_name: 'Mahindra Blazo',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Mahindra',
    performance: 8,
    timestamp: ISODate('2021-04-25T00:00:00.000Z'),
    price: 3000000
  }
]
```

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

```
Atlas atlas-8o45zw-shard-0 [primary] vehicles> db.two_wheelers.find({}, { _i_id: 0, bike_name: 1, price: 1 })
[
  { bike_name: 'Hero Splendor Plus', price: 65000 },
  { bike_name: 'Honda Activa 6G', price: 75000 },
  { bike_name: 'Bajaj Pulsar 150', price: 95000 },
  { bike_name: 'TVS Apache RTR 200', price: 135000 },
  { bike_name: 'Suzuki Access 125', price: 78000 }
]
Atlas atlas-8o45zw-shard-0 [primary] vehicles> db.four_wheelers.find({}, { __id: 0, vehicle_name: 1, price: 1 })
[
  { vehicle_name: 'Maruti Swift', price: 650000 },
  { vehicle_name: 'Tata Ace', price: 550000 },
  { vehicle_name: 'Ashok Leyland Boss', price: 2200000 },
  { vehicle_name: 'Volvo 9400', price: 8500000 },
  { vehicle_name: 'Mahindra Blazo', price: 3000000 }
]
```

8. Write a MongoDB query to display two_wheelers from a particular company

```
Atlas atlas-8o45zw-shard-0 [primary] vehicles> db.two_wheelers.find(
...   { manufacturer: "Honda" }
... ).pretty()
...
[
  {
    _id: ObjectId('69b13169ca81b07ee5718dc5'),
    bike_name: 'Honda Activa 6G',
    model: 'gearless',
    category: '125cc',
    colors_available: [ 'white', 'grey', 'blue' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: ISODate('2021-03-10T00:00:00.000Z'),
    price: 75000
  }
]
```

9. Write a MongoDB query to display four_wheelers available in diesel variants

```
Atlas atlas-8o45zw-shard-0 [primary] vehicles> db.four_wheelers.find(
...   { variants: { $regex: /diesel/i } }
... ).pretty()
...
[
  {
    _id: ObjectId('69b13170ca81b07ee5718dc9'),
    vehicle_name: 'Maruti Swift',
    model: 'own',
    category: 'car',
    variants: [ 'VXI petrol', 'ZXI petrol', 'VXI diesel' ],
    manufacturer: 'Maruti Suzuki',
    performance: 8,
    timestamp: ISODate('2018-05-10T00:00:00.000Z'),
    price: 650000
  },
```

```
  {
    _id: ObjectId('69b13170ca81b07ee5718dca'),
    vehicle_name: 'Tata Ace',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'diesel', 'CNG' ],
    manufacturer: 'Tata Motors',
    performance: 7,
    timestamp: ISODate('2017-09-01T00:00:00.000Z'),
    price: 550000
  },
  {
    _id: ObjectId('69b13170ca81b07ee5718dcb'),
    vehicle_name: 'Ashok Leyland Boss',
    model: 'commercial',
    category: 'lorry',
    variants: [ 'diesel' ],
    manufacturer: 'Ashok Leyland',
    performance: 9,
    timestamp: ISODate('2019-02-20T00:00:00.000Z'),
    price: 2200000
  },
  {
    _id: ObjectId('69b13170ca81b07ee5718dcc'),
    vehicle_name: 'Volvo 9400',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Volvo',
    performance: 9,
    timestamp: ISODate('2020-12-15T00:00:00.000Z'),
    price: 8500000
  },
  {
    _id: ObjectId('69b13170ca81b07ee5718dcd'),
    vehicle_name: 'Mahindra Blazo',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Mahindra',
    performance: 8,
```

```
    performance: 8,
    timestamp: ISODate('2021-04-25T00:00:00.000Z'),
    price: 3000000
  }
]
```

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

```
Atlas atlas-8o45zw-shard-0 [primary] vehicles> db.two_wheelers.find(
...   { performance: { $gt: 5 } },
...   { _id: 0, bike_name: 1, category: 1, manufacturer: 1 }
... )
[
  {
    bike_name: 'Hero Splendor Plus',
    category: '100cc',
    manufacturer: 'Hero MotoCorp'
  },
  {
    bike_name: 'Honda Activa 6G',
    category: '125cc',
    manufacturer: 'Honda'
  },
  {
    bike_name: 'Bajaj Pulsar 150',
    category: '150cc',
    manufacturer: 'Bajaj'
  },
  {
    bike_name: 'TVS Apache RTR 200',
    category: '200cc',
    manufacturer: 'TVS'
  },
  {
    bike_name: 'Suzuki Access 125',
    category: '125cc',
    manufacturer: 'Suzuki'
  }
]
```

```
Atlas atlas-8o45zw-shard-0 [primary] vehicles> db.four_wheelers.find(
...   { performance: { $gt: 5 } },
...   { _id: 0, vehicle_name: 1, category: 1, manufacturer: 1 }
... )
[
  {
    vehicle_name: 'Maruti Swift',
    category: 'car',
    manufacturer: 'Maruti Suzuki'
  },
  {
    vehicle_name: 'Tata Ace',
    category: 'mini truck',
    manufacturer: 'Tata Motors'
  },
  {
    vehicle_name: 'Ashok Leyland Boss',
    category: 'lorry',
    manufacturer: 'Ashok Leyland'
  },
  {
    vehicle_name: 'Volvo 9400',
    category: 'bus',
    manufacturer: 'Volvo'
  },
  {
    vehicle_name: 'Mahindra Blazo',
    category: 'heavy truck',
    manufacturer: 'Mahindra'
  }
]
```

2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.

```
Atlas atlas-8o45zw-shard-0 [primary] test> use animal
switched to db animal
```

2. Write a MongoDB query to display all the databases.

```
Atlas atlas-8o45zw-shard-0 [primary] animal> show dbs
sample_mflix  113.76 MiB
vehicles      144.00 KiB
admin         360.00 KiB
local          92.40 GiB
```

3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.

```
Atlas atlas-8o45zw-shard-0 [primary] animal> db.createCollection("wild_animals", {
...   capped: true,
...   size: 1048576,
...   max: 1000
... })
{ ok: 1 }
Atlas atlas-8o45zw-shard-0 [primary] animal> db.createCollection("domestic_animals")
{ ok: 1 }
```

4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```
Atlas atlas-8o45zw-shard-0 [primary] animal> // 4) Insert
... db.wild_animals.insertMany([
...   {
...     animal_name: "Lion",
...     nature: "harm",
...     favorite_foods: ["meat", "deer"],
...     care_taker_name: "Rahul",
...     life_span: 15,
...     timestamp: ISODate("2022-01-10T09:00:00Z"),
...     expenses: 50000
...   },
...   {
...     animal_name: "Tiger",
...     nature: "harm",
...     favorite_foods: ["meat", "boar"],
...     care_taker_name: "Rahul",
...     life_span: 18,
...     timestamp: ISODate("2021-11-20T10:30:00Z"),
...     expenses: 55000
...   },
...   {
...     animal_name: "Deer",
...     nature: "harmless",
...     favorite_foods: ["grass", "leaves"],
...     care_taker_name: "Anita",
...     life_span: 12,
...     timestamp: ISODate("2023-03-05T08:15:00Z"),
...     expenses: 20000
...   },
...   {
...     animal_name: "Elephant",
...     nature: "harmless",
...     favorite_foods: ["grass", "fruits"],
...     care_taker_name: "Suresh",
...     life_span: 60,
...     timestamp: ISODate("2020-07-18T14:45:00Z"),
...     expenses: 80000
...   },
```

```
...   {
...     animal_name: "Leopard",
...     nature: "harm",
...     favorite_foods: ["meat", "rabbits"],
...     care_taker_name: "Anita",
...     life_span: 17,
...     timestamp: ISODate("2022-09-25T12:00:00Z"),
...     expenses: 45000
...   }
... ])
...
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b135a670b2099773718dc4'),
    '1': ObjectId('69b135a670b2099773718dc5'),
    '2': ObjectId('69b135a670b2099773718dc6'),
    '3': ObjectId('69b135a670b2099773718dc7'),
    '4': ObjectId('69b135a670b2099773718dc8')
  }
}
```

5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```
Atlas atlas-8o45zw-shard-0 [primary] animal> // 5) Insert dome
... db.domestic_animals.insertMany([
...   {
...     animal_name: "Cow",
...     gender: "female",
...     favorite_foods: ["grass", "hay"],
...     animal_petname: "Gauri",
...     life_span: 20,
...     timestamp: ISODate("2021-02-14T10:00:00Z"),
...     expenses: 15000
...   },
...   {
...     animal_name: "Dog",
...     gender: "male",
...     favorite_foods: ["meat", "biscuits"],
...     animal_petname: "Bruno",
...     life_span: 13,
...     timestamp: ISODate("2022-05-01T09:30:00Z"),
...     expenses: 8000
...   },
...   {
...     animal_name: "Cat",
...     gender: "female",
...     favorite_foods: ["fish", "milk"],
...     animal_petname: "Kitty",
...     life_span: 14,
...     timestamp: ISODate("2023-01-22T16:20:00Z"),
...     expenses: 6000
...   },
...   {
...     animal_name: "Goat",
...     gender: "male",
...     favorite_foods: ["grass", "leaves"],
...     animal_petname: "Chintu",
...     life_span: 12,
...     timestamp: ISODate("2020-09-10T07:45:00Z"),
...     expenses: 7000
...   },
```

```
...    {
...      animal_name: "Buffalo",
...      gender: "female",
...      favorite_foods: ["grass", "fodder"],
...      animal_petname: "Rani",
...      life_span: 18,
...      timestamp: ISODate("2019-12-05T11:10:00Z"),
...      expenses: 14000
...    }
... ])
...
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b135ac70b2099773718dc9'),
    '1': ObjectId('69b135ac70b2099773718dca'),
    '2': ObjectId('69b135ac70b2099773718dcb'),
    '3': ObjectId('69b135ac70b2099773718dcc'),
    '4': ObjectId('69b135ac70b2099773718dcd')
  }
}
```

6. Write a MongoDB query to display all documents available in wild_animals and domastic_animals.

```
Atlas atlas-8o45zw-shard-0 [primary] animal> db.wild_animals.find().pretty()
[
  {
    _id: ObjectId('69b135a670b2099773718dc4'),
    animal_name: 'Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer' ],
    care_taker_name: 'Rahul',
    life_span: 15,
    timestamp: ISODate('2022-01-10T09:00:00.000Z'),
    expenses: 50000
  },
  {
    _id: ObjectId('69b135a670b2099773718dc5'),
    animal_name: 'Tiger',
    nature: 'harm',
    favorite_foods: [ 'meat', 'boar' ],
    care_taker_name: 'Rahul',
    life_span: 18,
    timestamp: ISODate('2021-11-20T10:30:00.000Z'),
    expenses: 55000
  },
  {
    _id: ObjectId('69b135a670b2099773718dc6'),
    animal_name: 'Deer',
    nature: 'harmless',
    favorite_foods: [ 'grass', 'leaves' ],
    care_taker_name: 'Anita',
    life_span: 12,
    timestamp: ISODate('2023-03-05T08:15:00.000Z'),
    expenses: 20000
  },
  {
    _id: ObjectId('69b135a670b2099773718dc7'),
    animal_name: 'Elephant',
    nature: 'harmless',
    favorite_foods: [ 'grass', 'fruits' ],
    care_taker_name: 'Suresh',
    life_span: 60,
    timestamp: ISODate('2020-07-18T14:45:00.000Z'),
```


```
    timestamp: ISODate('2020-07-18T14:45:00.000Z'),
```


```
    expenses: 80000
  },
  {
    _id: ObjectId('69b135a670b2099773718dc8'),
    animal_name: 'Leopard',
    nature: 'harm',
    favorite_foods: [ 'meat', 'rabbits' ],
    care_taker_name: 'Anita',
    life_span: 17,
    timestamp: ISODate('2022-09-25T12:00:00.000Z'),
    expenses: 45000
  }
]
```

```
Atlas atlas-8o45zw-shard-0 [primary] animal> db.domestic_animals.find().pretty()
[
  {
    _id: ObjectId('69b135ac70b2099773718dc9'),
    animal_name: 'Cow',
    gender: 'female',
    favorite_foods: [ 'grass', 'hay' ],
    animal_petname: 'Gauri',
    life_span: 20,
    timestamp: ISODate('2021-02-14T10:00:00.000Z'),
    expenses: 15000
  },
  {
    _id: ObjectId('69b135ac70b2099773718dca'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'biscuits' ],
    animal_petname: 'Bruno',
    life_span: 13,
    timestamp: ISODate('2022-05-01T09:30:00.000Z'),
    expenses: 8000
  },
  {
    _id: ObjectId('69b135ac70b2099773718dcb'),
    animal_name: 'Cat',
    gender: 'female',
    favorite_foods: [ 'fish', 'milk' ],
    animal_petname: 'Kitty',
    life_span: 14,
    timestamp: ISODate('2023-01-22T16:20:00.000Z'),
    expenses: 6000
  },
```

```
  {
    _id: ObjectId('69b135ac70b2099773718dcc'),
    animal_name: 'Goat',
    gender: 'male',
    favorite_foods: [ 'grass', 'leaves' ],
    animal_petname: 'Chintu',
    life_span: 12,
    timestamp: ISODate('2020-09-10T07:45:00.000Z'),
    expenses: 7000
  },
  {
    _id: ObjectId('69b135ac70b2099773718dcd'),
    animal_name: 'Buffalo',
    gender: 'female',
    favorite_foods: [ 'grass', 'fodder' ],
    animal_petname: 'Rani',
    life_span: 18,
    timestamp: ISODate('2019-12-05T11:10:00.000Z'),
    expenses: 14000
  }
]
```

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

```
Atlas atlas-8o45zw-shard-0 [primary] animal> db.wild_animals.find(
...   {},
...   { _id: 0, animal_name: 1, expenses: 1 }
... )
[
  { animal_name: 'Lion', expenses: 50000 },
  { animal_name: 'Tiger', expenses: 55000 },
  { animal_name: 'Deer', expenses: 20000 },
  { animal_name: 'Elephant', expenses: 80000 },
  { animal_name: 'Leopard', expenses: 45000 }
]
Atlas atlas-8o45zw-shard-0 [primary] animal> db.domestic_animals.find(
...   {},
...   { _id: 0, animal_name: 1, expenses: 1 }
... )
[
  { animal_name: 'Cow', expenses: 15000 },
  { animal_name: 'Dog', expenses: 8000 },
  { animal_name: 'Cat', expenses: 6000 },
  { animal_name: 'Goat', expenses: 7000 },
  { animal_name: 'Buffalo', expenses: 14000 }
]
```

8. Write a MongoDB query to display domestic_animals whose life is a particular year

```
Atlas atlas-8o45zw-shard-0 [primary] animal> db.domestic_animals.find(
...   { life_span: 12 }
... ).pretty()
[
  {
    _id: ObjectId('69b135ac70b2099773718dcc'),
    animal_name: 'Goat',
    gender: 'male',
    favorite_foods: [ 'grass', 'leaves' ],
    animal_petname: 'Chintu',
    life_span: 12,
    timestamp: ISODate('2020-09-10T07:45:00.000Z'),
    expenses: 7000
  }
]
```

9. Write a MongoDB query to display wild_animals available under a particular care_taker

```
Atlas atlas-8o45zw-shard-0 [primary] animal> db.wild_animals.find(
...   { care_taker_name: "Rahul" }
... ).pretty()
[
  {
    _id: ObjectId('69b135a670b2099773718dc4'),
    animal_name: 'Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer' ],
    care_taker_name: 'Rahul',
    life_span: 15,
    timestamp: ISODate('2022-01-10T09:00:00.000Z'),
    expenses: 50000
  },
  {
    _id: ObjectId('69b135a670b2099773718dc5'),
    animal_name: 'Tiger',
    nature: 'harm',
    favorite_foods: [ 'meat', 'boar' ],
    care_taker_name: 'Rahul',
    life_span: 18,
    timestamp: ISODate('2021-11-20T10:30:00.000Z'),
    expenses: 55000
  }
]
```

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

```
Atlas atlas-8o45zw-shard-0 [primary] animal> db.wild_animals.find(
...   { life_span: { $gt: 5 } },
...   { _id: 0, animal_name: 1, favorite_foods: 1, expenses: 1 }
... )
...
... // 10) From domestic_animals
... db.domestic_animals.find(
...   { life_span: { $gt: 5 } },
...   { _id: 0, animal_name: 1, favorite_foods: 1, expenses: 1 }
... )
[
  {
    animal_name: 'Cow',
    favorite_foods: [ 'grass', 'hay' ],
    expenses: 15000
  },
  {
    animal_name: 'Dog',
    favorite_foods: [ 'meat', 'biscuits' ],
    expenses: 8000
  },
  {
    animal_name: 'Cat',
    favorite_foods: [ 'fish', 'milk' ],
    expenses: 6000
  },
  {
    animal_name: 'Goat',
    favorite_foods: [ 'grass', 'leaves' ],
    expenses: 7000
  },
  {
    animal_name: 'Buffalo',
    favorite_foods: [ 'grass', 'fodder' ],
    expenses: 14000
```

```
[
  {
    animal_name: 'Lion',
    favorite_foods: [ 'meat', 'deer' ],
    expenses: 50000
  },
  {
    animal_name: 'Tiger',
    favorite_foods: [ 'meat', 'boar' ],
    expenses: 55000
  },
  {
    animal_name: 'Deer',
    favorite_foods: [ 'grass', 'leaves' ],
    expenses: 20000
  },
  {
    animal_name: 'Elephant',
    favorite_foods: [ 'grass', 'fruits' ],
    expenses: 80000
  },
  {
    animal_name: 'Leopard',
    favorite_foods: [ 'meat', 'rabbits' ],
    expenses: 45000
  }
]
```